ASRC Fuel Tank Robot Daily Report

Separate rope lanes, lane-routed 3D modules and a complete build package, 13 November 2026

Until today the model routed all three rope sets through the same four holes at 24 mm, which three 2 mm ropes cannot share. Today each set got its own lane, the whole simulation pipeline was re-run with it, and the 3D model was renewed to match: lane holes, rope anchors, a backbone hole sized to the 1.8 mm rod, module joints, end parts and a step-by-step build guide.

Summary of today's work

- **Rope lanes (radial spread).** Set A (anchored at U4) at 21 mm, B (M5) at 24 mm, C (M6) at 27 mm, on the same up/right/down/left rays. Startup, yaw sweep, reference round trip and closed loop were re-run. Everything passes; peak rope force falls from 17.4 to 16.4 N and the smallest stability margin rises from 0.166 to 0.185.
- **Rope-path limit found.** Ropes are guided only at the end disks, so inside a bent module each rope is a straight chord. M5 bends 14 deg at the 0 deg hold but up to 103 deg when steering; beyond about 15 deg yaw its ropes leave the strip-free corridor (6.7-7.8 mm) and would rub the M spring strips.
- **3D model renewed** (cad/lanes_u4m2). The STEP spring strips lie exactly on the rope rays, so the bodies were re-clocked (U -45 deg, M -22.5 deg). Solid end disks keep only working holes; Ø1.85 mm backbone bore with 15 mm grip per end; rope anchors; joint holes.
- **Build package.** Base interface and head adapter added; spring pockets, dowel holes and rope-A anchors corrected; an automatic whole-assembly check passes (all pass); BUILD_GUIDE.md covers printing to tensioning.
- **Repository.** The lane simulation moved to its own folder, simulations/u4m2/lanes_20261113, reusing the solver in backbone_20261108 unchanged.

Check (3 N pretension, 20 deg)	Shared 24 mm holes	Lanes 21/24/27 mm
Startup, 32 states	all pass, peak 17.4 N	all pass, peak 16.3 N
Yaw holds 0-175 deg (36 poses)	peak 17.4 N, min eig 0.166	all pass, peak 16.4 N, min eig 0.185
Reference round trip	all pass, peak 17.6 N, clearance 20.6 mm	all pass (1101 states), peak 16.5 N, clearance 20.1 mm
Closed loop, 5 cases, settled samples	all pass	all pass
Head 550 g, states above 1 deg while moving	115 (max 3.13 deg)	48 (max 2.32 deg)

Decisions from today
Adopt separate rope lanes (A 21, B 24, C 27 mm) and the re-clocked, lane-routed modules as the current design. Limit first hardware tests to plus or minus 15 deg yaw until rope guides are added inside M5/M6 and modelled. Build from cad/lanes_u4m2/BUILD_GUIDE.md.

Simulation with separate rope lanes

lanes.py gives every anchor set its own rope offset inside a copy of the pinned spatial kernel; the pinned files are not changed. With all radii at 24 mm it reproduces the pinned kernel to 1e-14 (energy) and 1e-16 (gradient); with lanes, the finite-difference gradient error is 2e-8. lanes_study.py then runs the unchanged pipeline (startup, minimum-force yaw sweep, reference round trip, controller gains, closed loop) with the lanes installed.

Case	Shared holes v3: angle / tension (settled)	Lanes: angle / tension (settled)	Lanes: failed checks
Nominal	0.18 deg / 2.98-17.7 N	0.14 deg / 2.99-16.6 N	0
Rod E 75 GPa	0.20 deg / 2.92-22.5 N	0.20 deg / 2.90-22.5 N	0
Head 550 g	0.20 deg / 2.76-23.7 N	0.67 deg / 2.78-20.8 N	0
Free rope 1.5 m	0.20 deg / 2.81-18.0 N	0.20 deg / 2.81-16.9 N	0
Rope EA -20%	0.20 deg / 2.89-17.9 N	0.20 deg / 2.89-16.8 N	0

Closed loop with gains rebuilt from the lane reference, no correction during moves. Angle = largest yaw or elevation error of a settled sample.

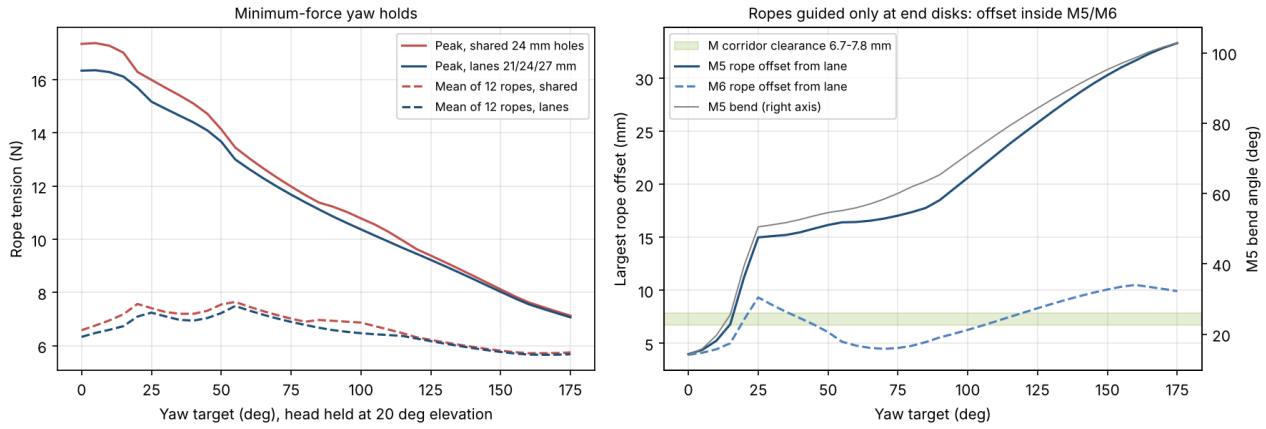


Figure 1 Left: peak and mean rope tension of the 20 deg yaw holds, shared holes (red) and lanes (blue). Right: largest offset of a straight rope from its lane inside M5 and M6, against the corridor clearance of the re-clocked M body (green), and the M5 bend angle.

- **Lanes help slightly everywhere.** Peak tensions are about 1 N lower in the holds and 1-3 N lower in closed loop (the 75 GPa case is unchanged). The 550 g head's moving error at 174-178 deg yaw shrinks from 115 to 48 states above 1 deg (max 3.1 to 2.3 deg) but does not vanish; online gain updating remains the fix.
- **Rope chords.** U modules bend at most 7.3 deg and their ropes stay within 0.2 mm of the lane. M5 rope offset: 4.0 mm at the 0 deg hold, 6.8 mm at 15 deg, 15 mm at 25 deg and 33 mm at 175 deg. The shared-hole layout has the same problem. Rod strain at the largest bend is about 1.7%.

Renewed 3D model: lane-routed modules

Built by cad/lanes_u4m2/build_lane_modules.py from the STEP-derived meshes, which stay untouched.

- **Re-clocking was required.** In both STEP bodies the spring strips lie on the 0/90/180/270 deg rays, where the ropes must run: without it every lane passes through material. Rotating U by -45 deg and M by -22.5 deg gives every straight rope 4.1-16.6 mm of clearance, and puts U's stiff leaf-spring direction vertical, as the simulation assumes.
- **Solid end disks.** Every 3 mm end disk is a full round plate; notches, strip-root relief slots, U diagonal slots and the unused U centre hole are filled. Left: 12 rope holes (2.5 mm, only for sets that still pass), 4 joint holes, U leaf-spring pockets, M backbone bore.
- **Backbone hole matches the rod.** M5/M6: Ø1.85 mm bore for the Ø1.8 mm NiTi rod, with a 6 mm sleeve inside each end disk giving 15 mm of grip per end, so the rod is clamped in rotation and bends with the module.
- **Rope anchors.** With 3 mm between lanes a crimp cannot sit on a lane. Each rope loops through a groove on its anchor disk's outer face and back down a return hole, and is crimped on the inside: B and C loop sideways (5 mm), A loops inward (3.5 mm) to stay clear of the U4 leaf springs.
- **Joints.** Rigid disk-to-disk: 2 dowel pins (left/right) set the clocking so every rope hole lines up, 2 M2 screws (up/down) clamp, nuts inside the module.

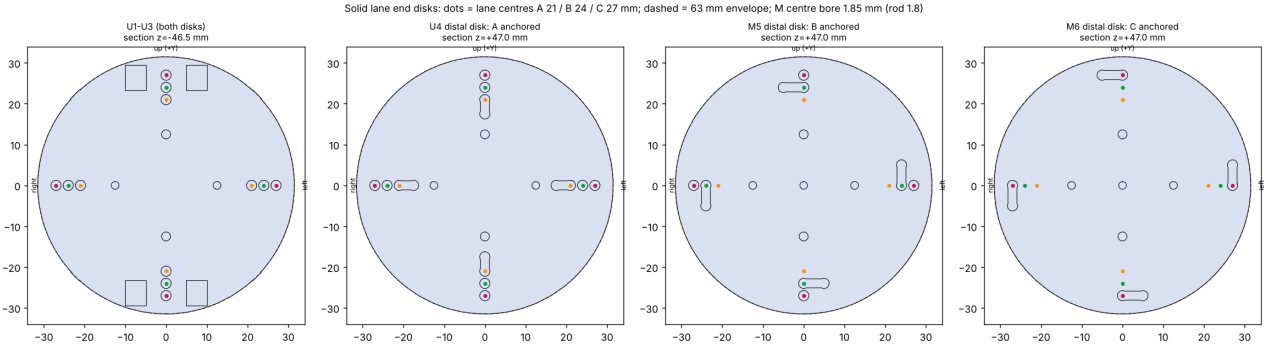


Figure 2 End disks: U1-U3, U4 distal (A anchors), M5 distal (B anchors), M6 distal (C anchors). Dots: lane centres A 21 / B 24 / C 27 mm.

Build package and assembly check

Part	Qty	Volume, cm3	Role
base_interface	1	60.1	20 mm block on the rear beam; guides all 12 ropes into U1; 4 x M3 inserts
U_lane_pass	3	23.9	U1-U3
U4_lane_anchorA	1	25.1	U4, anchors rope set A
M5_lane_anchorB	1	31.4	M5, anchors set B, rod bore
M6_lane_anchorC	1	31.6	M6, anchors set C, rod bore
head_adapter	1	18.3	6 mm plate: closes the C anchors, rod-end relief, 4 x M3 inserts for the head

- **Purchased:** 16 NiTi leaf-spring strips about 6.0 x 1.1 x 94 mm; one Ø1.8 x 192 mm NiTi rod; 12 lengths of Ø2 mm PA6 monofilament (1.51 / 1.60 / 1.70 m for A / B / C); 12 crimp sleeves; 14 Ø2 x 8 mm dowels; 14 M2 x 10 screws and nuts; 8 M3 heat-set inserts; epoxy.
- **Corrections made while writing the guide:** leaf-spring pockets re-opened (through on the proximal disk so the strip slides in, blind on the distal disk); dowel holes made Ø1.9 pilots for reaming (a Ø2 pin in the Ø2.2 screw hole would not locate); rope A's anchor turned inward (a sideways block came within 0.3 mm of the U4 leaf springs); hole chamfers reduced so a wall stays between lanes at the 3 mm pitch.

Automatic check (build_ends_and_check.py)	Result
7 joints: 4 joint holes open and aligned	pass
12 ropes: clear path base to anchor, groove, return hole, no hole beyond	pass
Rod: bore open through M5 and M6, U4 centre solid	pass
Leaf springs shifted 1.5 mm: no contact; M2 nut fits at every joint hole	pass
Largest radius of any part	31.66 mm (same as the STEP bodies)

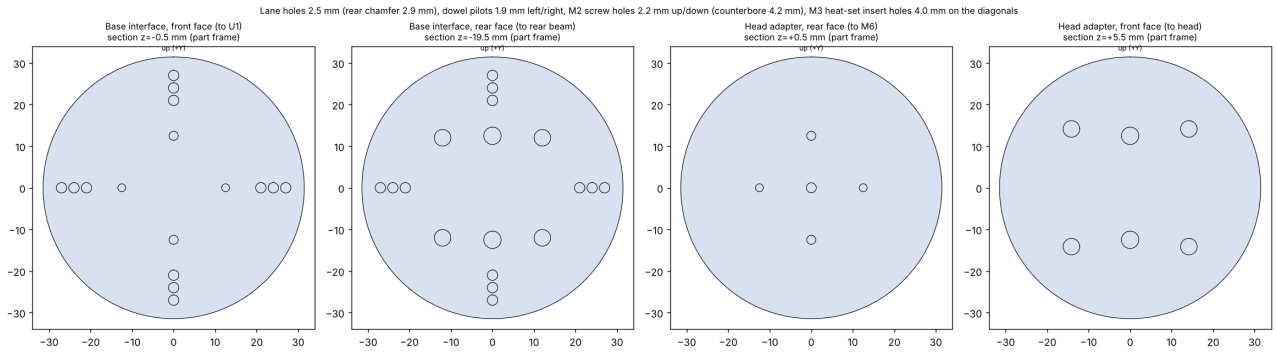


Figure 3 Base interface (front and rear faces) and head adapter (rear and front faces).

Limitations

- Ropes are modelled as straight chords between end disks. Results beyond about 15 deg yaw assume rope paths the current M body cannot give without intermediate guides.
- The CAD is checked geometrically only, in the straight pose. Printed strength (0.5 mm walls between rope holes, anchor bridges), rope friction and the stiffness of the re-clocked bodies with filled disk slots are untested.
- The printed parts are heavier than the model: M5/M6 about +5 g each, U +1.3 g each, plus about 1.3 g of joint hardware per joint; the head adapter (17 g) counts toward the 500 g head. The simulation has not been re-run with these masses.
- Quasi-static model with ideal measurements; the five closed-loop cases were used during development and are not an independent validation set.
- No wiring path through the modules; base and head mounting patterns are placeholders.

Next steps

Priority	Task	Purpose
1	Rope guides in M5/M6	Add intermediate guides and route the ropes through them in the model (per-cell chords) to allow steering beyond 15 deg yaw
2	Update masses, re-run	Use the printed part masses and joint hardware in the model; repeat the lane pipeline
3	Online gain update	Remove the remaining 550 g large-yaw transient
4	Print and test	One U and one M module: stiffness vs model, 0.5 mm walls, anchor at 50 N, rope friction (BUILD_GUIDE.md section 10)
5	Interfaces	Real rear-beam and head mounting patterns; head wiring path

Scripts and data: simulations/u4m2/lanes_20261113 (lanes.py, lanes_study.py, rope_chord_check.py, lanes_figures.py, README); cad/lanes_u4m2 (build_lane_modules.py, build_ends_and_check.py, plot_disks.py, BUILD_GUIDE.md, README). This report is generated by reports/builders/build_report_2026_11_13.py.